EXPERIMENT : 3

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1.COMMAND LINE INTERFACE:

PYTHON CODE:

```
records = {}
def insert_record():
    name = input("Enter name: ")
    detail = input("Enter detail: ")
    records[name] = detail
    print("Record inserted successfully!")

def update_record():
    name = input("Enter name to update: ")

    if name in records:
        new_detail = input("Enter new detail: ")
        records[name] = new_detail
        print("Record updated successfully!")
    else:
        print("Record not found!")

while True:
```

```
    print("\n1. Insert Record")
    print("2. Update Record")
    print("3. Display Records")
    print("4. Exit")

    choice = input("Enter choice: ")

    if choice == "1":
        insert_record()
    elif choice == "2":
        update_record()
    elif choice == "3":
        print(records)
    elif choice == "4":
        break
    else:
        print("Invalid choice")
```

OUTPUT:

```
PS C:\Users\SABITHA R\uiux> py -3.12 "cli.py"

1. Insert Record
2. Update Record
3. Display Records
4. Exit
Enter choice: 1
Enter name: sabitha
Enter detail: 18
Record inserted successfully!
```

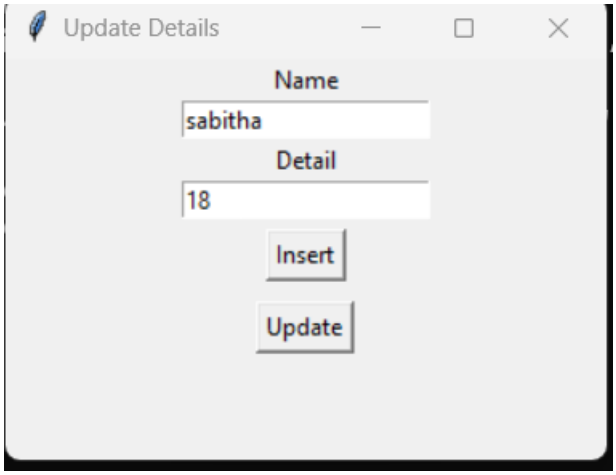
2.GRAPICAL USER INTERFACE:

PYTHON CODE:

```
import tkinter as tk
from tkinter import messagebox

records = {}

def insert_record():
    name = name_entry.get().strip()
    detail = detail_entry.get().strip()

    if name == "" or detail == "":
        messagebox.showwarning("Warning", "Enter all details")
        return

    records[name] = detail
    messagebox.showinfo("Success", "Record Inserted")

def update_record():
    name = name_entry.get().strip()

    if name in records:
        new_detail = detail_entry.get().strip()
        records[name] = new_detail
        messagebox.showinfo("Success", "Record Updated")
```

```
    else:
        messagebox.showerror("Error", "Record not found")

root = tk.Tk()
root.title("Update Details")
root.geometry("300x200")

tk.Label(root, text="Name").pack()
name_entry = tk.Entry(root)
name_entry.pack()

tk.Label(root, text="Detail").pack()
detail_entry = tk.Entry(root)
detail_entry.pack()

tk.Button(root, text="Insert", command=insert_record).pack(pady=5)
tk.Button(root, text="Update", command=update_record).pack(pady=5)

root.mainloop()
```

OUTPUT :



3.VOICE USER INTERFACE:

PYTHON CODE:

```
import speech_recognition as sr

records = {}

def process_command(command):
    command = command.lower()

    if "insert" in command:
        parts = command.split(" ")



    name = parts[1]
        detail = parts[2]
        records[name] = detail
```

```
        print("Record inserted")

    elif "update" in command:
        parts = command.split(" ")
        name = parts[1]
        new_detail = parts[2]

        if name in records:
            records[name] = new_detail
            print("Record updated")
        else:
            print("Record not found")

def listen():
    r = sr.Recognizer()

    with sr.Microphone() as source:
        print("Speak command...")
        audio = r.listen(source)

    try:
        command = r.recognize_google(audio)



  print("Command:", command)
        process_command(command)
```

```
    except:
        print("Could not understand")

while True:
    listen()
```

OUTPUT:

```
PS C:\Users\SABITHA R> cd uiux
PS C:\Users\SABITHA R\uiux> py -3.12 "vui.py"
Speak command...
Could not understand
```

4.USER SATISFACTION COMPARISON:

PYTHON CODE:

```
def survey():
    print("Rate satisfaction (1-5)")

    try:
        cli = int(input("CLI: "))
        gui = int(input("GUI: "))

   vui = int(input("VUI: "))

        avg = (cli + gui + vui) / 3

        print("\nAverage Satisfaction:", avg)
```

```
    except:
        print("Invalid Input")

survey()
```

OUTPUT:

```
PS C:\Users\SABITHA R> cd uiux
PS C:\Users\SABITHA R\uiux> py -3.12 "satis.py"
Rate satisfaction (1-5)
CLI: 2
GUI: 3
VUI: 2

Average Satisfaction: 2.3333333333333335
PS C:\Users\SABITHA R\uiux>
```